



Commvault configuration

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1. Architecture Overview

1.1 Core Components

- **CommServe**
 - Central management and metadata server
 - No backup data stored here
- **Media Agents / Access Nodes (VMs)**
 - Deployed as **VMs per PCD region**
 - Responsible for:
 - Data movement (backup & restore)
 - Hosting local disk backup storage
 - Acting as restore proxies

1.2 OS Separation

Component	Windows	Linux
Backup	Either MA can move data	Either MA can move data
Restore	Windows MA required	Linux MA required

2. CommServe Installation

2.1 Requirements

- Supported OS (Windows or Linux)
- As per Commvault system requirements

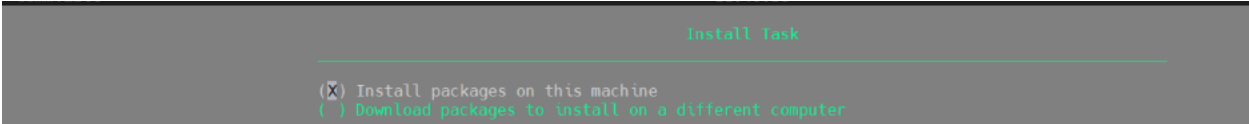
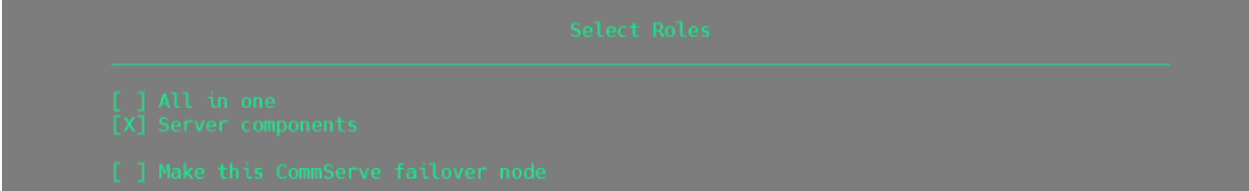
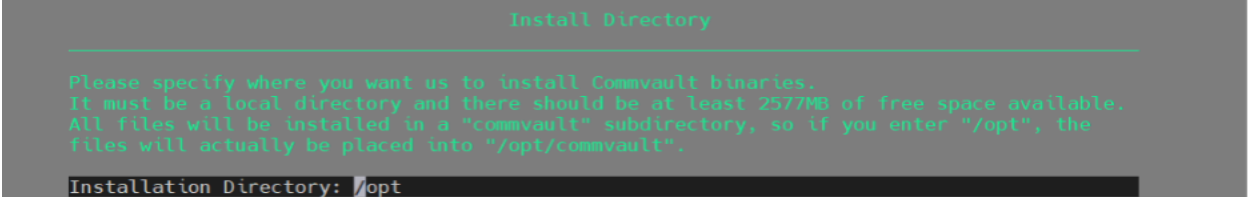
https://documentation.commvault.com/11.20/system_and_hardware_requirements_for_commvault.html

2.2 Installation

1. Download Commvault Media Kit
2. Run installer:

```
[root@Commvault-srv ~]# ll
total 36308
-rw-----, 1 root root      909 Dec 22 09:46 anaconda-ks.cfg
-rwxrwxrwx, 1 root root 18584396 Aug 15 10:49 Commvault_linux-x8664_Media_11_40
-rw-r--r--, 1 root root 18586112 Jan  4 08:33 Commvault_linux-x8664_Media_11_40.tar
[root@Commvault-srv ~]# ./Commvault_linux-x8664_Media_11_40
Using bootstrapper installer /root/Commvault_linux-x8664_Media_11_40:v1181222
Using repository https://downloadcenter.commvault.com
- Downloading get_iflist4.....
```

3. Choose:
 - Install required packages locally
 - Configure **CommCell**
 - i. Installing required packages either on the server itself or on a different computer (will go with the first option)
 - ii. There are many options depending on what you want to configure (in this case commcell)

- 
- 
- 
- 
- 

4. Proceed with default setup

```
Commvault Database Setup Recommended Settings

We recommend the following settings. To customize, remove the selections for one or more settings.

[X] Use "Commvault-srv" as the hostname
[X] Use "Commvault-srv" as the client name
[X] Use "/var/log/commvault/Log_Files" as the log directory
[X] Use "/opt/commvaultdata" as the data directory
[X] Use "/opt/DM2CacheDir" as the web cache directory
[X] Use "/opt/commvaultDB" as the database directory
[X] Create new database instead of restoring from database dump
[X] Use "/opt/commvaultDR" as the disaster recovery directory
```

5. Enter the Commvault credentials. These credentials will be used to access the Commvault Command Center via the web interface.

```
Commvault ID

Please enter your Commvault ID account.

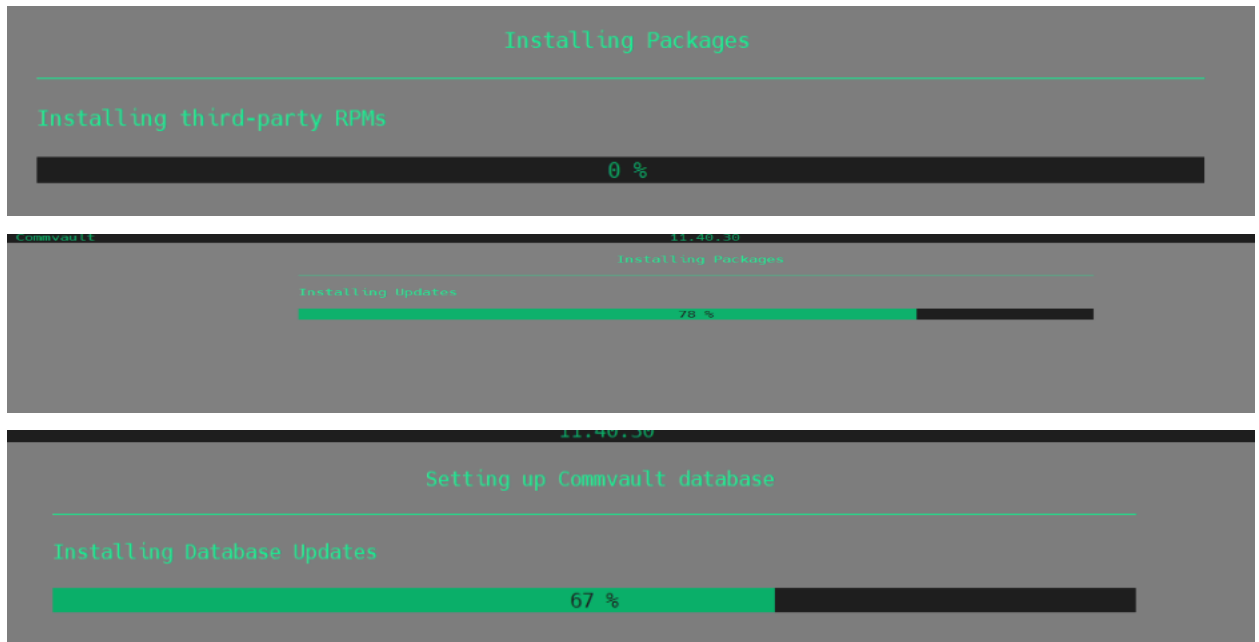
User Name: admin
Email: admin@admin.local
Password: *****
Confirm Password: *****
```

6. Wait for package download & installation (can take ~1 hour)

```
Summary

Please review the following before installation

Instance           : Instance001
Roles groups selected : Server components
Packages selected   : Metrics Server
                    : Command Center
                    : CommServe
                    : WebServer
                    : WorkFlow Engine
                    : MongoDB
                    : ManagedService
Dependent packages  : File System Core
                    : Message Queue
                    : TaskManager Service
                    : File System
Installation Directory : /opt/commvault
Log Directory         : /var/log/commvault/Log_Files
Maintenance release   : 30
```

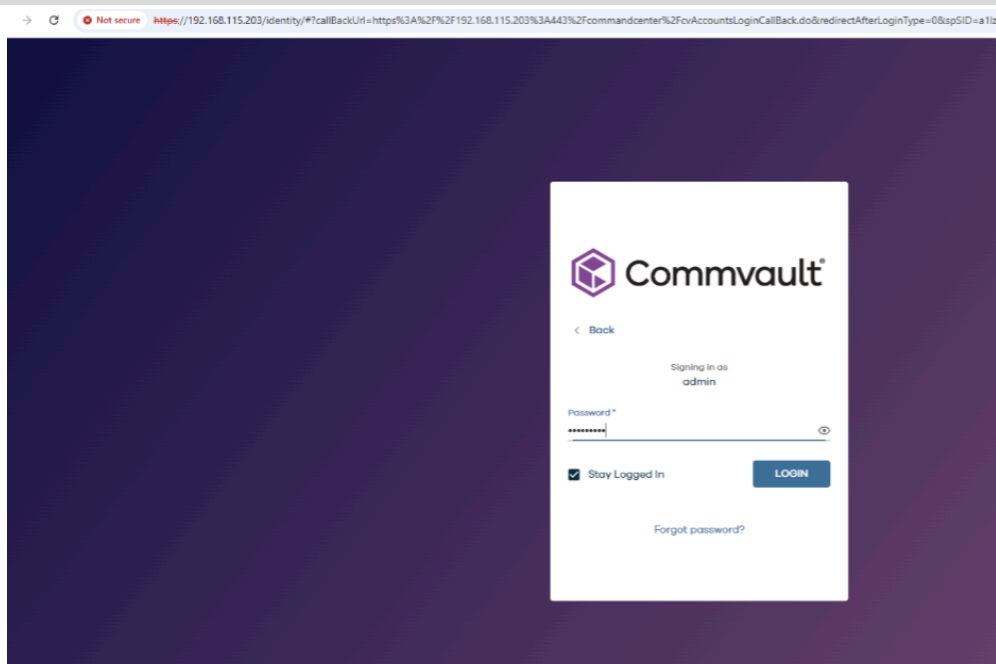


2.3 Post-Install

- Commvault services running

Access UI:

<https://<CommServe-IP>>



3. Backup Storage Configuration

3.1 Primary Storage (Recommended)

Disk

Add disk storage

Name *

Backup location Add

No backup location added

Use deduplication ☒

Deduplication DB location Add

No deduplication db location added

EQUIVALENT API

CANCEL SAVE

- **Local disk storage on Media Agent VMs**
- Disk Library configuration:
 - High IOPS
 - Sized based on retention & workload
- Benefits:
 - Faster restores (low RTO)
 - No WAN dependency during recovery

3.2 Secondary Storage (Optional – Later Phase)

Cloud

Add cloud storage

Type

Microsoft Azure Storage

HP E-Class Storage

Huawei Object Storage

IBM Cloud Object Storage (S3-compatible)

Microsoft Azure Storage

NetApp StorageGRID

Nutanix Objects

OpenStack Object Storage

Oracle Cloud Infrastructure Archive Storage

Oracle Cloud Infrastructure Object Storage

Pure Storage FlashBlade

IAM AD application

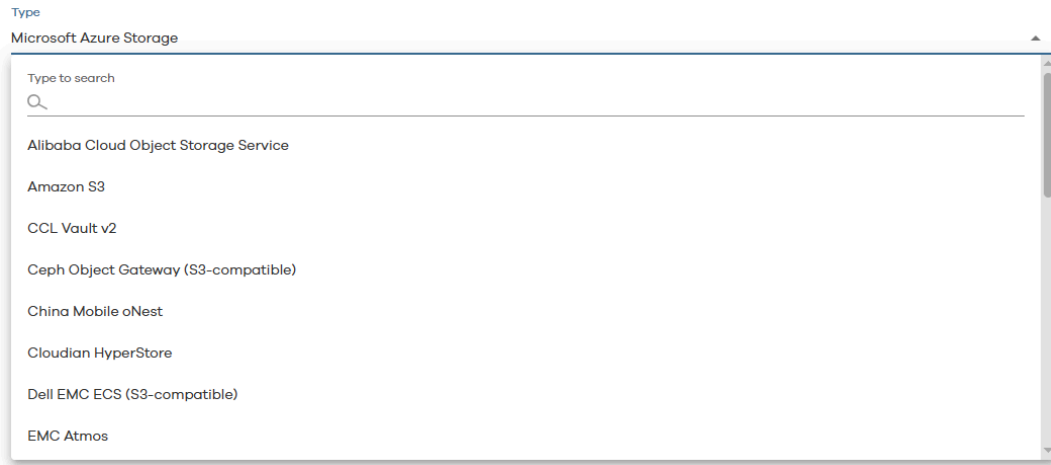
Account name *

Credentials *

Click to select

Container *

Use deduplication ☒



- Cloud storage (S3 / Azure / etc.)
- Used for:
 - Secondary copy
 - Long-term retention
 - DR use cases

4. Media Agent / Access Node Deployment

Vms can be configured to do various jobs depends on the packages installed on it :

☐ Advanced Indexing Pack	☐ INFORMIX
☐ Block Level Replication	☐ Media Agent
☐ Cloud Apps	☐ MongoDB
☐ Command Center	☐ MySQL
☐ Content Analyzer	☐ NoSQL
☐ Content Extractor	☐ OFFICE365
☐ DB2	☐ Oracle
☐ Domino Database	☐ PostgreSQL

General Rules

- Deployed as **VMs in PCD**
- One **Windows MA** + one **Linux MA** per region
[VSA requirements](#)
- Local disks attached for backup storage

Create a Custom Installation Package

1. Execute the Commvault installer script on the CommServe server.

```

Welcome to Commvault Installer

-----

For support please visit http://www.commvault.com/support, or send an email to
support@commvault.com, or call US/CANADA (Toll Free) (877) 780-3077; INTERNATIONAL (732)
571-2160

This installer will install Commvault packages on this machine. Log Messages will be sent
to /var/log/.gxsetup/logs/20260104/install.log

```

2. Select Download package to install on a different computer.

```

11:40:30

Install Task

-----

( ) Install packages on this machine
(X) Download packages to install on a different computer

```

3. Choose the correct media option (e.g., Linux / Unix Custom Package).

```

Download Media Options

-----

(X) Download and create Linux/UNIX custom package
( ) Download and create Linux/UNIX native package
( ) Download and create Windows custom package
( ) Download UNIX OS appliance updates

```

4. Select the correct operating system platform

```

Platform Selection

-----

Please choose one or more platforms to package from the list below.

[ ] AIX-PPC
[ ] FreeBSD-x86
[ ] FreeBSD-x86-64
[ ] HP-IA64
[ ] Linux-ARM64
[ ] Linux-PPC
[ ] Linux-PPC64-LE
[ ] Linux-S390
[ ] Linux-x86
[X] Linux-x86-64
[ ] macOS-64
[ ] macOS-ARM64
[ ] Solaris-Sparc
[ ] Solaris-x86-64

```


5. Choose Advanced to select specific components.

```
Download Media Options
-----
( ) Full media kit
(X) Advanced (package-based selection)
```

6. Select Join an existing CommCell.

```
Install Option
-----
Please select one of the following:
( ) Media Kit - Create a new CommCell
(X) Media Kit - Join an existing CommCell
( ) Advanced Selection
```

7. Select the required roles.

```
Select Roles
-----
[X] Virtualization
[X] File server
[ ] Cloud apps
[ ] Databases
[ ] Laptop
[X] MediaAgent
[ ] Big data
[ ] Data Insights
[ ] Gateways
[ ] CommServe failover
[ ] Threat Scan
```

8. Verify the package directory.

```
Custom Package Directory
-----
Please enter the directory name where custom package has to be created.
Custom Package Directory: /opt
```

9. Select the Latest Maintenance Release.

```
11:46:30
Select Maintenance Release
-----
(X) Latest Maintenance Release
( ) Specify Maintenance Release

[ ] Include custom releases
```

10. Review the configuration and proceed.

```
11.40.30
Summary
-----
Please review the following

Download media option : Advanced (package-based selection)

Roles groups selected : Virtualization
                      File server
                      MediaAgent

Packages selected    : MediaAgent
                      Virtual Server
                      Remote Plugin Server
                      File System
                      Cloud Apps
                      IntelliSnap Legacy

Dependent packages   : File System Core
                      Storage Accelerator

Installation Directory : /opt/DownloadPackageLocation/Unix

Maintenance release  : 30
```

11. Wait while the custom package is created.

```
11.40.30
Creating custom package
-----
Downloading files
[ ] 3 %
```

12. Record the custom package answers for reference.

```
Record Answers
-----
Do you want to record the answers for custom package install?
( X ) Yes
(   ) No
```

13. Leave the Instance Number empty (use the current instance).

```
11.40.30
Instance Page
-----
Please specify instance number (1-20). If no input is provided, a new instance is always
created.
Instance:
[ ] Show to user
```

14. Join an existing CommCell

```
11:40:30
Install Option
-----
Please select one of the following:
(X) Join an existing CommCell
( ) Advanced Selection
```

15. Select the requirement roles

```
11:40:30
Select Roles
-----
[X] Virtualization
[X] File server
[ ] Cloud apps
[ ] Laptop
[X] MediaAgent
[ ] Gateways
[ ] Show to user
```

16. Select No for restore only agent

```
Install Agents for Restore Only
-----
Do you want to install the agents for restore only without consuming licenses?
( ) Yes
(X) No
[ ] Show to user
```

17. Check the installation directory

```
11:40:30
Install Directory
-----
Please specify where you want us to install Commvault binaries.
It must be a local directory or NFS and there should be at least 298MB of free space
available. All files will be installed in a "commvault" subdirectory, so if you enter
"/opt", the files will actually be placed into "/opt/commvault".

Installation Directory: /opt
[ ] Show to user
```

18. Select No for dedicated Unix group

```
Unix Group Assignment
-----
Would you like to assign a dedicated Unix group?
( ) Yes
(X) No
[ ] Show to user
```

19. Leave the CVD Port empty to use the default

```
CVD/Network Port

Every instance of Commvault should use a unique set of network ports to avoid interfering
with other instances running on the same machine

Please enter CVD port number:

[ ] Show to user
```

20. Set the Data Directory to /opt

```
Data Directory

The Data directory is designated for I/O intensive operations such as deduplication and
index caching. To optimize performance and simplify storage capacity management, it is
recommended to use local disk storage pools for the Data directory whenever possible.

Please specify the location for the Data Directory. All files will be saved in the
commvaultdata subdirectory. For example, if you enter "/opt", the files will be placed in
"/opt/commvaultdata".

Data Directory: /opt

[ ] Show to user
```

21. Review the configuration before starting installation

```
Summary

Please review the following before installation

Instance           : New
Roles groups selected : Virtualization
                    : File server
                    : MediaAgent
Packages selected   : MediaAgent
                    : Virtual Server
                    : Remote Plugin Server
                    : File System
                    : Cloud Apps
                    : IntelliSnap Legacy
Dependent packages   : File System Core
                    : Storage Accelerator
Installation Directory : /opt
Log Directory         : /var/log

[ ] Show to user
```

22. Select No for Server pre-client certificate

```
Server Per-Client Certificate

If you selected "Force per-client certificate authentication on CommServe" on the server
or gateway, then you must provide the installer with a temp certificate.
To generate a temp certificate, access the Control Panel in the CommCell Console, open the
Certificate Administration Applet, click the Temp Certificate button, select the newly
created client in the popped up window, and then copy or paste the certificate that is
generated to a local file that can be accessed by the installer.

Did you enable per-client certificate authentication on the server?

( ) Yes
(X) No

[ ] Show to user
```

23. Provide the CommServe hostname or IP address

```
Server Information
-----
Enter CommServe or Gateway hostname. For port number other than 8403, use the format
hostname:port.

CommServe/Gateway hostname: comm-srv

[ ] CommServe will connect to this computer to finish registration.

[ ] Specify if there is a HTTP proxy between this client and the server (for example,
    Squid or Apache.)

[ ] Post Quantum Cryptography (PQC) mode enabled on Commserve

[ ] Show to user
```

24. Select No for Decoupled install

```
Decoupled Install
-----
Do you want to treat the install as a decoupled install?

( ) Yes
(X) No

[ ] Show to user
```

25. Select No for Ignore name conflict

```
Override Client Info
-----
Do you want to ignore client/host name conflict on CommServe?

( ) Yes
(X) No

[ ] Show to user
```

26. Enter the CommServe authentication credentials

```
Server Authentication
-----
To register the client, you must enter a valid username and password, an authorization
code, a QSDK Access Token, or a subscription ID.

Enter credentials

Username: admin
Password: *****

or, enter Authorization Code

AuthCode:

[ ] Show to user
```

27. Leave the plan option empty for now

```
Plans
-----
Please provide a plan name
Plan:
[ ] Show to user
```

28. Client group No

```
11.40.30
Configure the Client Groups
-----
Please provide a client group:
Do you want to add another client group?
( ) Yes
(X) No
[ ] Show to user
```

29. Subclient Policy Empty

```
11.40.30
Configure the Subclient Policy
-----
Please provide a subclient policy:
[ ] Show to user
```

30. Leave the Storage policy empty for now

```
11.40.30
Configure the Storage Policy
-----
Please provide a storage policy:
[ ] Show to user
```

31. Prepare tar file Yes

```
11.40.30
Tar File Creation
-----
Optionally, you may choose to tar the custom package in a tar file now.
Do you want to create the tar file now?
(X) Yes
( ) No
```

The custom package (TAR file) will be created under:

/opt/DownloadPackageLocation/tar

```

                                Tar File Creation
-----
Optionally, you may choose to tar the custom package in a tar file now.
Do you want to create the tar file now?

(X) Yes
( ) No

```

5. Linux Media Agent (Access Node)

5.1 OS-Level Requirements

Install the following packages (required for restore & file browse):

- QEMU disk image utility (qemu-img)
- libguestfs
- libguestfs-tools
- Logical Volume Management (lvm)

https://documentation.commvault.com/v11/commcell-console/system_requirements_for_virtual_server_agent_with_openstack_03.html#virtual-server-agent-proxy-requirements

5.2 Required Commvault Packages

Installed from Commvault UI:

- Linux File System
- Virtual Server Agent
- MediaAgent

5.3 Installation Steps

Access nodes can be configured either through the UI or using the way in the previous chapter “**Create a Custom Installation Package** “

1. Go to **Commvault UI**
2. **Manage** → **Servers** → **Add Server**
3. Select Linux server

4. Push install required packages
5. Verify Media Agent appears under **Servers**

Add server

☒

Install software packages on your computer

☐

Complete the installation if software packages are already installed on your computer.

Host name *

OS Type

☒ Windows ☐ Unix and Linux

Use saved credentials ☐

User name *
Example: domain\username

Password *

Confirm password *

Select package(s) *
Click to select

Installation location

Restore only agent ☐

EQUIVALENT API

CANCELINSTALL

6. Windows Media Agent (Access Node)

6.1 OS-Level Requirements

- Install **.NET Framework 4.8**
- Required before Commvault package installation

6.2 Installation Method (Important)

Windows clients **cannot** be installed via Linux CommServe UI for some versions.

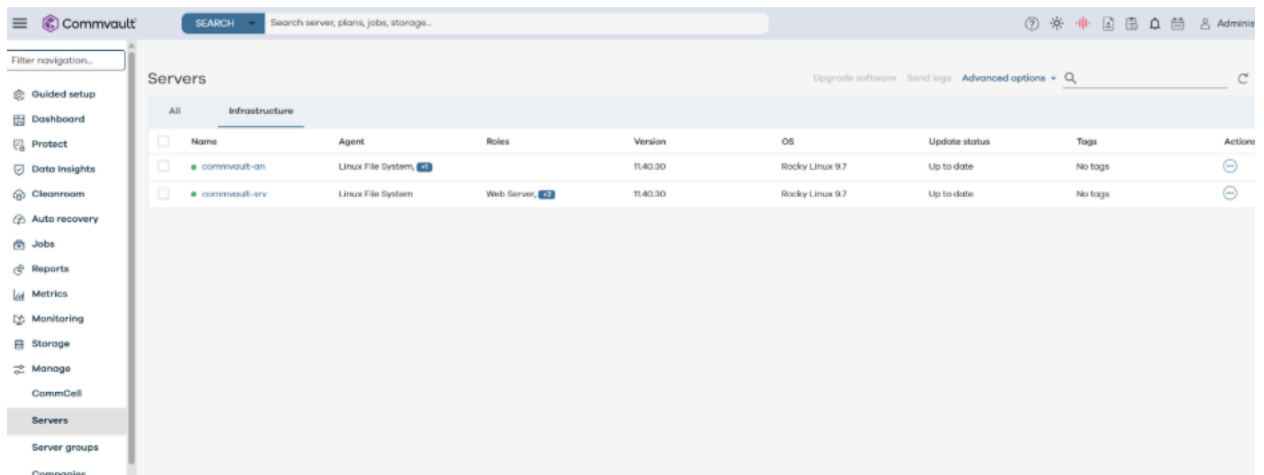
Correct approach:

1. Copy the generated TAR file to the Access Node server.
2. Execute the silent installation script (silent_install).

```
[root@commvault-an Unix]# ll
total 380
-rwxrwx---. 1 root root 132 Jan 4 06:42 cvpkgadd
-rwxrwx---. 1 root root 52 Jan 4 06:42 cvpkgseed
-rw-r--r--. 1 root root 4561 Jan 4 06:58 default.xml
-rwxrwx---. 1 root root 87025 Jan 4 06:45 detect
drwxrwx---. 4 root root 111 Jan 4 06:45 linux-x8664
-rwxrwx---. 1 root root 45 Jan 4 06:45 pkgs_inmedia.txt
-rwxrwx---. 1 root root 129612 Jan 4 06:38 pkg.xml
-rwxrwx---. 1 root root 10 Jan 4 06:38 pkg.xml.description
-rwxrwx---. 1 root root 129612 Jan 4 06:38 rompkg.xml
-rwxrwx---. 1 root root 132 Jan 4 06:42 silent_install
-rwxrwx---. 1 root root 52 Jan 4 06:42 support
-rwxrwx---. 1 root root 141 Jan 4 06:38 version
[root@commvault-an Unix]# ./silent_install
Using update installer /root/.gxsetup/silent_install/newinstall:v1181589
```

6.3 Verification

- Appears as Media Agent in UI
- Assigned as Windows restore proxy



7. Adding Platform9 Cloud Director (PCD)

1. Go to **Virtualization** → **Add Hypervisor**
2. Select **OpenStack**

Configure Hypervisor

Select vendor

☐ Amazon Web Services 	☐ Microsoft Azure 	☐ Google Cloud Platform 	☐ Nutanix AHV 
☐ Microsoft Hyper-V 	☐ VMware vCenter 	☐ Alibaba Cloud 	☐ Microsoft Azure Stack Hub 
☐ OpenStack 	☐ Oracle Cloud Infrastructure 	☐ Oracle VM 	☐ Red Hat Virtualization 
☐ VMware Cloud Director 	☐ Xen 	☐ Fusion Compute 	☐ Proxmox VE 

3. Enter:

- PCD Keystone endpoint (add the port 443)
- Add domain user as **Default**
- Credentials
- Project / Region details

Hypervisors /

Configure OpenStack Hypervisor

1 Add Hypervisor

2 Add VM Group

3 Summary

Add Hypervisor

Keystone Address *

Hypervisor display name *

OpenStack Domain User

User name *

Password *

Access nodes *

Click to select

EQUIVALENT API

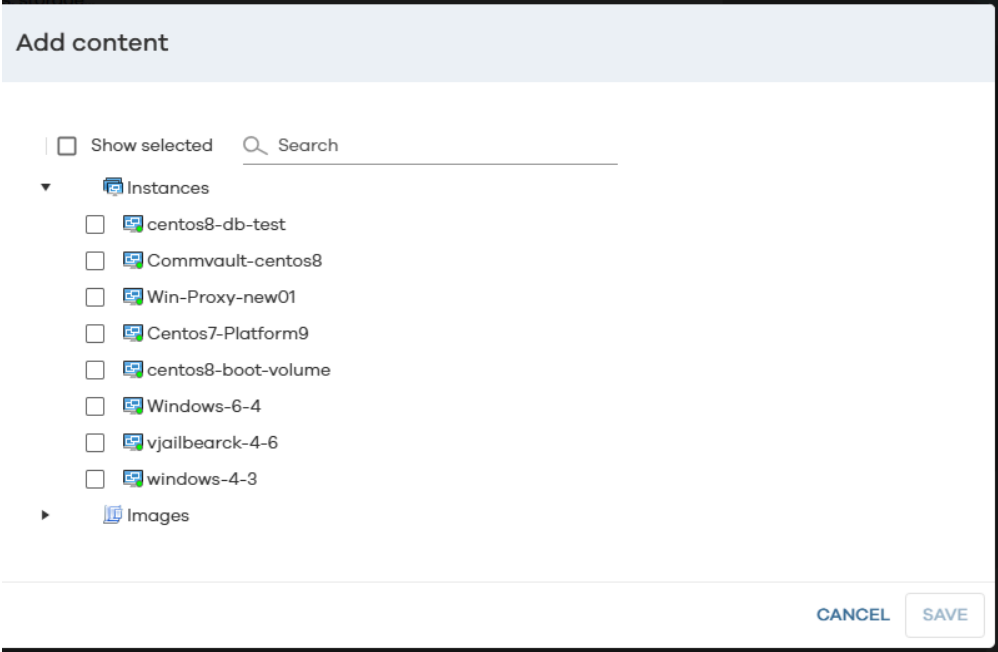
CANCEL NEXT

4. Validate connection
5. Discover PCD VMs

8. Adding VMs for Backup

VM Groups

- Create VM Groups using:
 - Content (explicit VMs)
 - Rules (tags / naming patterns)



The screenshot shows a dialog box titled "Add content". At the top, there is a header bar with the text "Add content". Below this, there is a search bar with a magnifying glass icon and the text "Search". To the left of the search bar is a checkbox labeled "Show selected". Below the search bar, there is a list of items under the heading "Instances". The items are: "centos8-db-test", "Commvault-centos8", "Win-Proxy-new01", "Centos7-Platform9", "centos8-boot-volume", "Windows-6-4", "vjailbearck-4-6", and "windows-4-3". Each item has a small icon to its left and a checkbox to its right. Below the "Instances" list, there is a heading "Images" with a right-pointing arrow next to it. At the bottom right of the dialog box, there are two buttons: "CANCEL" and "SAVE".

Backup Configuration

- Assign:
 - Storage Policy
 - Media Agent
- Schedule backups

9. Restore Behavior (Important for Design)

Backup Phase

- Media Agent OS **does not need to match** VM OS
- Data movement only

Restore Phase

- OS-specific proxy **is mandatory**
- Examples:
 - Windows VM restore → **Windows Media Agent**
 - Linux VM restore → **Linux Media Agent**
- Reason:
 - File systems
 - Guest tools
 - OS-specific restore operations

10. Database Protection (Optional)

Database protection is **agent-based**.

1. Add DB server:
 - **Manage → Servers → Add Server**
2. Install **directly on the same database server**:
 - File System Agent
 - Application-specific agent (MySQL, MSSQL, etc.)
3. Configure DB backup policies
4. Verify visibility under **Database** tab

Overview

Instances

Databases

Instant clones

Add instance

Add server

Instances

1 Met

0 Missed

3 Excluded

All

Type = All

+ Add filter

Name ↑ 2	Server ↑ 1	Database engine	Status	Last backup	Application size	Plans	SLA status	Actions
centos8-boot-volume_3306	centos8-boot-volume	MySQL	Not Ready	Apr 8, 2025, 4:04 P...	1MB	PF9-TestPlan	Excluded	
centos8-db-test_3306	centos8-db-test	MySQL	Ready	Apr 14, 2025, 12:02 ...	1MB	PF9-TestPlan	Excluded	

Since the agent is installed **on the database server itself**, backups are performed in an **agent-based, application-aware manner**, ensuring consistent backups and supported restore operations.

11. Backup and Restore Workflow in Platform9 Cloud Director (PCD)

This section explains the sequence of operations and the resources created in **PCD** when a **backup** or **restore** job is triggered from Commvault.

1. Backup Workflow

High-Level Flow

Commvault interacts with PCD using the **OpenStack APIs**. Backups are **agentless at the VM level** (for VMs), with data movement handled by the **Media Agent (Access Node)**.

Sequence of Operations

1. Backup Job Triggered

- A backup job is initiated from Commvault (scheduled or manual).
- The job is associated with a VM Group and a Storage Policy.

2. PCD / OpenStack Authentication

- Commvault authenticates to PCD via **Keystone** using the configured credentials.
- Project and region context is established.

3. VM Metadata Collection

- Commvault queries PCD for:
 - VM details
 - Attached volumes
 - Disk layout
 - Snapshot capabilities

4. Snapshot Creation (PCD Side)

- PCD creates:
 - A **VM snapshot** or
 - **Volume snapshots** (depending on configuration)
- **Note: Only VMs that are booted from persistent (bootable) volumes are supported for backup and restore.**

VMs created with ephemeral disks booted directly from images are not protected by Commvault.

- The VM remains powered on unless otherwise configured.

5. Temporary Snapshot Resources Created

On the PCD side, the following resources are created temporarily:

- Snapshot objects (VM or volume)
- Temporary snapshot volumes
- Snapshot metadata entries

6. Data Read by Media Agent

- The Media Agent (Access Node VM) accesses the snapshot data.
- A **temporary volume created from the snapshot is attached or mapped to the Media Agent VM**, allowing it to read the data without impacting the source VM.
- Backup data is streamed:
 - From snapshot → Media Agent
 - From Media Agent → local disk library

7. Snapshot Cleanup

- Once data transfer completes successfully:
 - Snapshot resources are **deleted**
 - Temporary volumes are **removed**
- PCD returns the VM to its original state.

8. Job Completion

- Backup metadata is updated in Commvault.
- Backup data is retained according to the Storage Policy.

PCD Impact Summary (Backup)

- No permanent resources left behind
- Short-lived snapshots only
- No agent installed inside the VM
- Minimal VM performance impact

2. Restore Workflow

High-Level Flow

Restores are **proxy-based** and require an **OS-matching Media Agent**. Commvault uses PCD APIs to recreate or overwrite VM resources.

Sequence of Operations

1. Restore Job Triggered

- Restore initiated from Commvault:
 - Full VM restore
 - File-level restore

2. Restore Proxy Selection

- Commvault selects an **OS-specific Media Agent**:
 - Windows VM restore → Windows Media Agent
 - Linux VM restore → Linux Media Agent

3. PCD Authentication

- Commvault authenticates to PCD using Keystone.
- Target project, network, cluster and region are selected.

4. Target Resource Preparation (PCD Side)

Depending on restore type, PCD creates:

- New volumes from backup data
- Temporary volumes (during restore)
- Network attachments

5. Data Transfer

- Backup data is read from:
 - Local disk library on Media Agent
- Data is written to:
 - Newly created or existing PCD volumes

6. VM Reconstruction

- For full VM restore:
 - VM definition is recreated
 - Volumes attached
 - Networks connected
 - VM powered on (optional)

7. Cleanup of Temporary Resources

- Temporary restore volumes or staging resources are removed.
- Final VM and disks remain active.

8. Restore Job Completion

- Restore status updated in Commvault.
- VM becomes available in PCD.

3. PCD Impact Summary (Restore)

- New resources **may be permanently created**:
 - VMs
 - Volumes
 - Network attachments
- Temporary resources are cleaned automatically
- Restore speed depends on:
 - Media Agent proximity
 - Local disk performance
 - Volume size

4. Key Design Notes (Client-Relevant)

- Backups are **snapshot-based and agentless** for VMs.
- Database backups remain **agent-based** (inside the DB server).
- Media Agents handle all data movement.
- Local disk storage ensures:
 - Faster restores
 - Reduced dependency on network or cloud storage
- OS-specific Media Agents are **mandatory for restores**, not backups.

12. Summary (Client-Facing Key Points)

- CommServe = control plane only
- Media Agents = data plane + storage
- Local disk storage on Media Agents
- OS-specific proxies required **for restores**, not backups
- Deployed per PCD region for performance & isolation